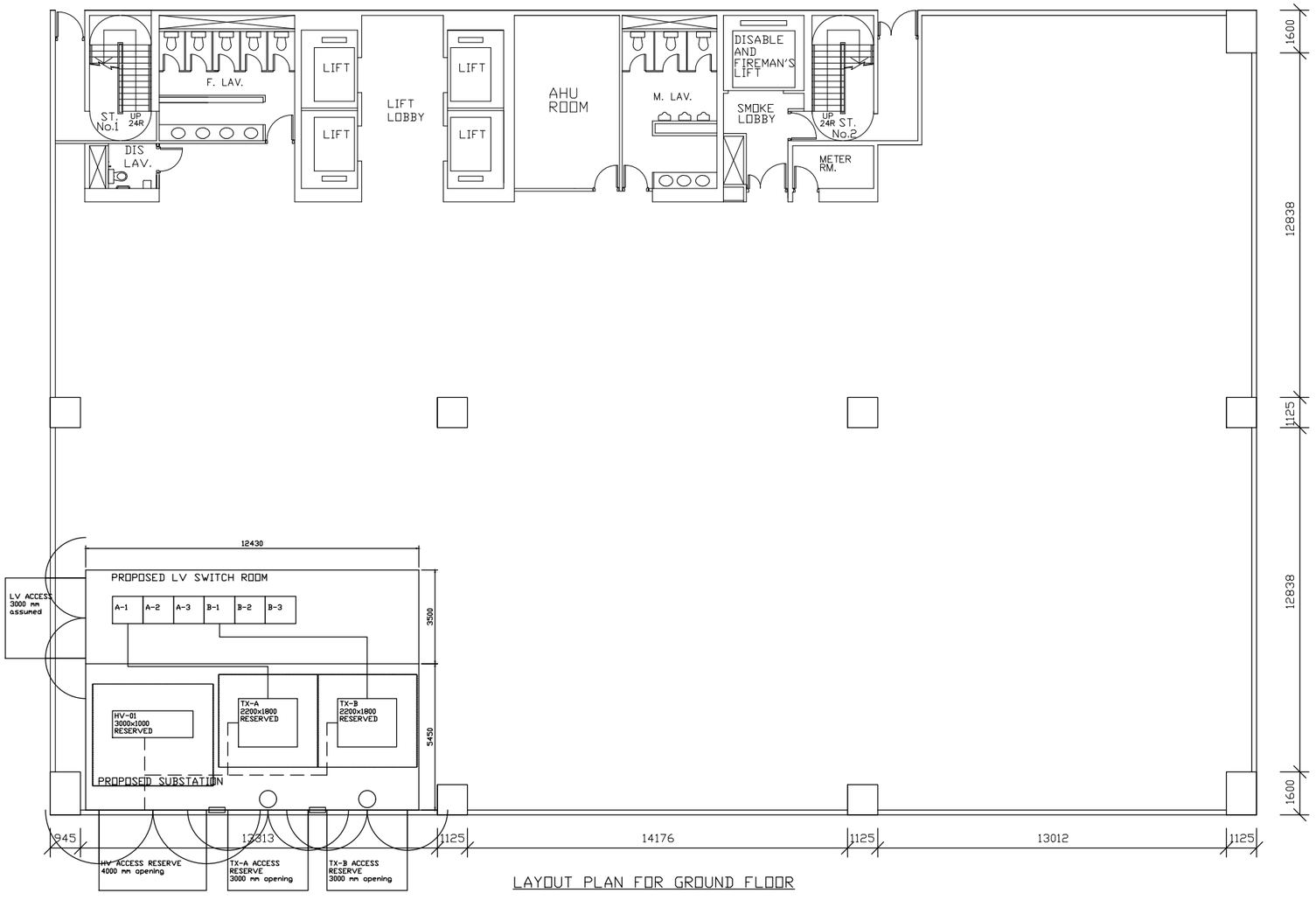
TASK 1 | GROUND-FLOOR PLANTROOM LOCATION

PROPOSAL ON SUPPLIED ARCHITECTURAL PLAN | COURSE DESIGN REVIEW | PLAN 1:200 AT A3



PROPOSED LOCATION

Lower-left perimeter of supplied ground-floor plan. Substation 12430 x 5450; adjacent LV room 12430 x 3500 assumed. See separate 1:75 room detail.

LOCATION REASONS

Direct external plant access can be reserved subject to site confirmation.

Adjacent LV room keeps transformer-to-incomer interfaces short.

Perimeter plantrooms separate supply equipment from the upper lift/stair/AHU core and remaining accommodation.

LV room has its own proposed west entrance.

LOAD ASSESSMENT

Offices: 1644.800 kVA.

AC: 1000 kW cooling -> 526.039 kW electrical -> 584.488 kVA at assumed PF 0.90.

Public services: 200 kVA aggregate allowance.

Two-zone conservative envelope: 2672.216 kVA.

Propose 2 x 1500 kVA, with equal zone allocation; no full-load N-1.

SITE CONDITIONS TO VERIFY

External apron, road and <=20m unloading route are not given by the source plan. Maintain >=3m clear delivery route.

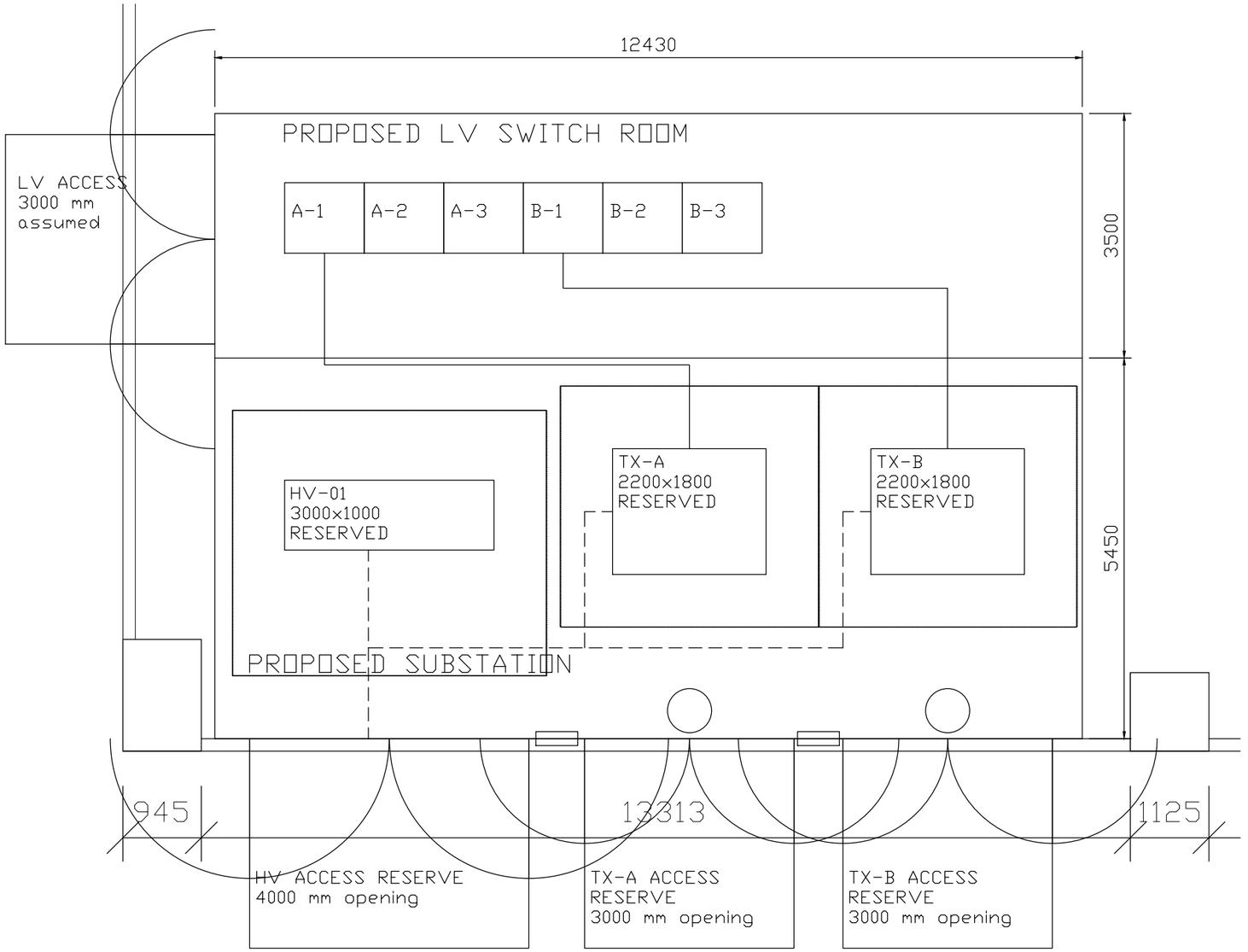
Architectural wall openings, structural/flood/fire provisions, ventilation and supplier equipment approval require coordination.

DRAWING REFERENCES: assignment_working_v02.dwg | TASK1-GROUND-OVERVIEW + TASK1-ROOM-REVIEW

Calculation sources and declared assumptions: Task1_answer_draft.md; data/task1_design_inputs.csv. Original drawings retained.

TASK 1 | PROPOSED SUBSTATION AND MAIN LV ROOM

Office building: 10 typical floors + ground | COURSE DESIGN REVIEW | All dimensions in mm



LINE KEY: dashed = HV trench; solid = LV route; chain = clearance envelope.

LV INTERFACES / HV ROUTES
LV trench: 1000W x 1100D, four MCTs per interface. A: 8.03+2=10.03m; B: 8.31+2=10.31m (<20m). 2m is assumed total termination/vertical allowance.
HV centre-lines: 1000W at panels; 600W branches; final 300W; generally 1400D. Segregate HV/LV. Covered trenches; no beam obstruction.

DETAIL 1:75 AT A3 | NOT A CONSTRUCTION DRAWING

CAPACITY AND EQUIPMENT
2 x 1500 kVA proposed; normal load envelope 2672.216 kVA. No full-load N-1.
HV: 3000 x 1000 reservation.
TX-A/B: 2200 x 1800 each.
These are assumed equipment envelopes, not manufacturer footprints.

CLEARANCES / BUILDING WORKS
HV rear 1000, front 1800, sides 750. TX LV terminals north: 900; other sides 750.
Proposed 2h FRR partitions; fire-seal penetrations. FFL +0.15 vs external +0.00 assumed.
Actual structures, wall thickness and equipment dimensions require coordination.

ACCESS AND DOORS
South external access assumed. Reserve 3000-wide delivery route; unloading distance <=20m to verify.
Plant doors: HV 4000W; TX 3000W each; 2800H assumed. 180-degree outward sweep, shown open 90 degrees.
Shared sweeps: use adjacent doors sequentially. TX-A designated wicket personnel access. Independent LV entrance west.

VENTILATION
2 low-level 600W x 2000H inlet louvres: 2.4m2 gross x assumed 50% free area = 1.20m2 >=1.12m2. Certified free area >=46.67% required.
2 high-level 630-dia fan symbols are conditional references. Actual losses, fan duty and noise to confirm; outlet >=2.5m above street, away from people.

Basis: supplied COP215 Rev07; COP101 v16 clauses 4.3, 5.5, 5.6, 6.1-6.7.
See Task1_answer_draft.md and shared data for assumptions/calculations.